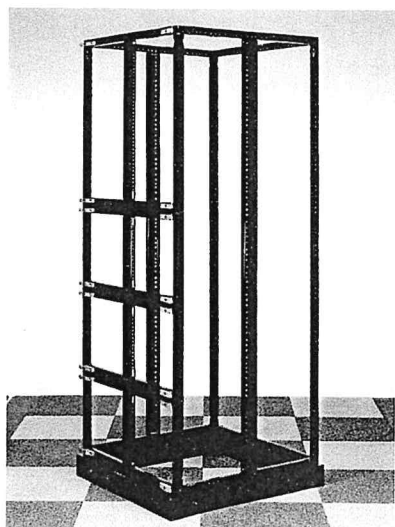
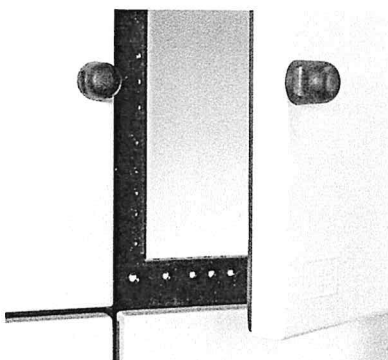


BASIC FRAME

基本框架



Base frame arrangement [4 modules x 4 modules x 10 modules].
基本框架 (4 模数 x 4 模数 x 10 模数)。



Panel door with gasket door hinge can be placed in the left or right.
小室门带执圈，铰链可左右安装。

ABB MD 190 (HONOR) LOW VOLTAGE SWITCHBOARD is tested in accordance with TILVA and ASTA requirements and built to BS and IEC Standards. The design is based on 190mm modular system and is therefore suitable for extension, rearrangement of the system at your own will.

FRAME — The basic 190mm modular frames (HF, HM) are made from 2mm thick electrogalvanised steel sheet which are coupled to a unique corner joining the three axis (X, Y, Z).

DOOR AND PANEL — All doors, panel covers and side plates are made of 1.5mm thick (or 2mm thick on request) electrogalvanised steel plate which are inter-changeable. The panel covers and side plates can be built with or without louvers or with ventilation fans for cooling and can be installed at any side of the switchboard. The concealed hinged doors may be arranged in the front, back and both sides of the switchboards. The doors are fitted with integral gasket and earth stud as standard production practice.

BASE FRAME — The base frame are made of 3mm thick electrogalvanised steel plate finished with black colour. The standard height of the base frame is 95mm and is suitable for flat floor or above cable trench installation.

FINISHING COLOUR — The whole switchboard are finished with stoved textured flint grey epoxy powder coating to RAL 7032 or RAL 7035 and other alternative colours upon request.

ABB MD 190 (HONOR) ‘安亚’ 系列 低压开关柜，是经过TILVA和ASTA试验，并符合BS、IEC标准。组件模数为190mm。开关柜结构可按用户要求扩展设备。

框架

190mm 模数组件 (HF, HM) 是用 2mm 厚电镀钢板制成，组件并与三轴 (X、Y、Z) 的单向面相接。

门及板盖

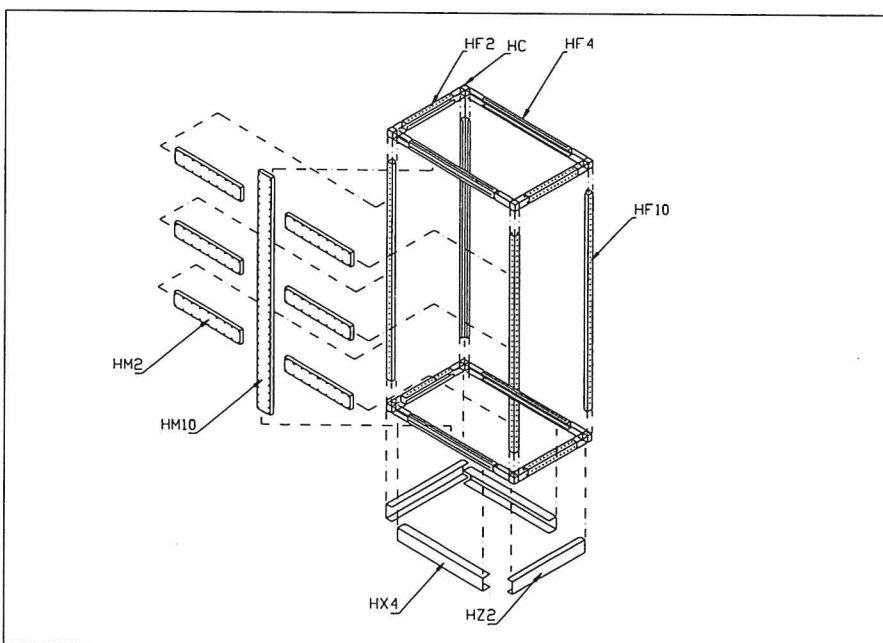
所有门、顶板和侧板都采用 1.5mm 厚 (或可供 2mm) 电镀钢板。另部件可互换。侧板如有需要可另要求有通风孔和无通风孔两种，在开关柜的任何一边安装冷却用风机。铰链门可分别安装在柜正面、背面和侧面。门板上配有垫圈和接地栓。

底框架

基本框架是用 3mm 厚电镀钢板并漆成黑色。其标准高度为 95mm。

颜色

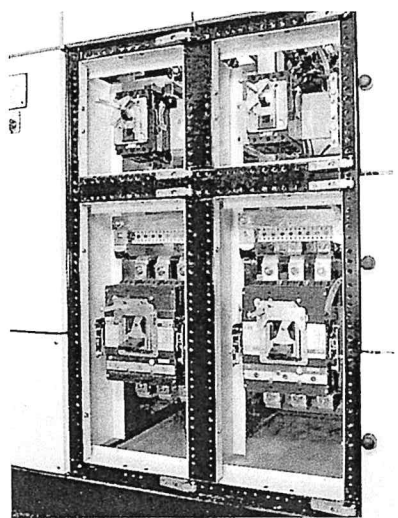
整个开关柜可用环氧烘粉涂成浅灰色 RAL 7032 (ABB 标准颜色) 或 RAL 7035 及其它颜色可按要求提供。



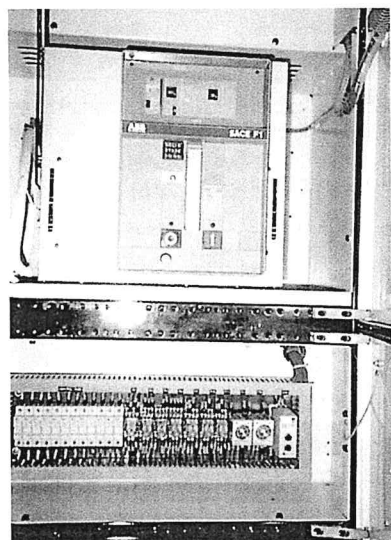
Assembling of [2 modules x 4 modules x 10 modules] cubicle.
(2 模数 x 4 模数 x 10 模数) 的组装图。

SWITCHBOARD DESIGN

开关柜设计



250A and 800A MCCB installation in the panel.
250A 及 800A MCCB 开关小室。



Provide suitable switchgear and control compartment.
2500A ACB 开关小室及控制小室。

ABB MD 190 (HONOR) Switchboard may be arranged as either front access type or rear access type. The switchboard is generally subdivided into the following individual compartments:

1. SWITCHGEAR COMPARTMENT
2. BUSBAR COMPARTMENT
3. CABLE COMPARTMENT

The compartments are made of 1.5mm thick electrogalvanised steel plate which is finished with light grey to RAL 7032 or orange colour for mounting plate.

Height and depth of the standard rear access type are 1995mm (10 modules + base frame 95mm) or 2185mm (11 modules + base frame 95mm) and 1010mm (5 modules + 60mm) respectively. Other optional combinations are available.

For the front access type, the standard height and depth are 1995mm or 2185mm and 820mm (4 modules + 60mm) respectively. Other optional dimensions are also available upon request.

ABB MD 190 (HONOR) '安亚' 系列 开关柜布置成前出线式或后出线式。开关柜分为下列各小室：

1. 开关小室
2. 母线小室
3. 电缆小室

所有的小室均采用 1.5mm 电镀锌钢板，并且用环氧烘粉涂成灰色 (RAL 7032) 或橙色。

后出线式：高度为 1995mm (10 模数 + 基架 95mm) 或 2185mm (11 模数 + 基架 95mm)，或其他理想模数组合，而深度为 1010mm (5 模数 + 60mm) 或其他模数组合。

前出线式：标准高度为 1995mm 或 2185mm，而深度为 820mm (4 模数 + 60mm)。其它尺寸也可按要求制造。

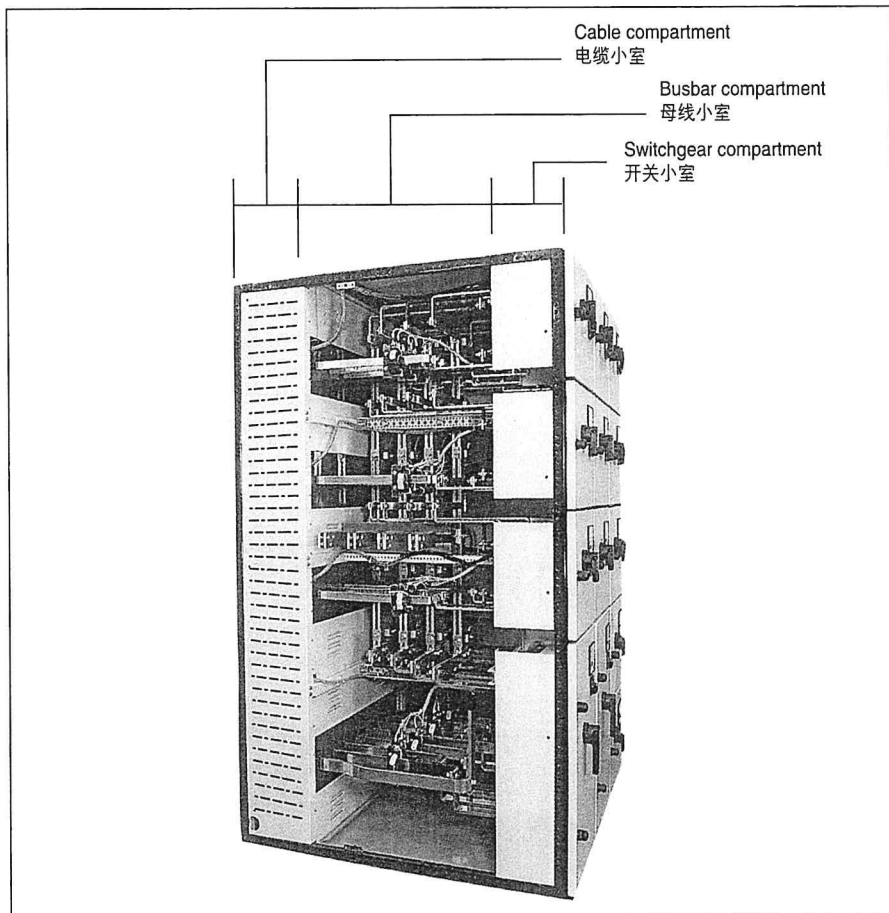
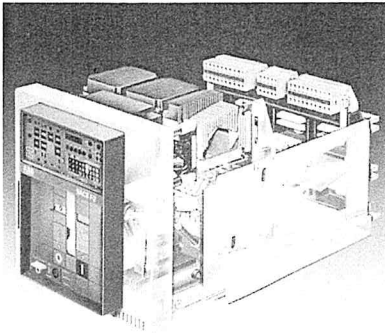


ABB MD190 (HONOR) switchboard side view showing clear the switchgear compartments, busbar compartments and cable compartments. (Standard Form 3b design)

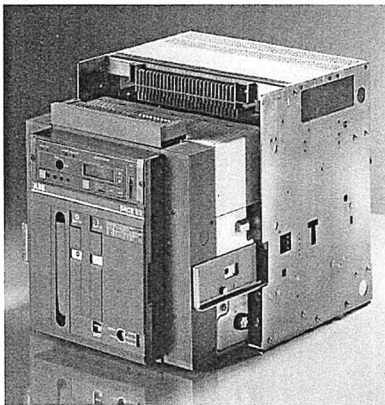
ABB MD190 (HONOR) '安亚' 开关柜侧面图，清楚地分为开关小室、母线小室及电缆小室。(型式 3b 设计)

SWITCHGEAR COMPARTMENT

开关柜小室



'ABB SACE' - Megamax ACB
ABB SACE - Megamax F 系列空气断路器



'ABB SACE' - Emax ACB
ABB SACE - Emax 空气断路器

All switchgears, meters and functional devices are housed inside the compartment.

SWITCHGEARS:

- AIR CIRCUIT BREAKER (ABB SACE) UP TO 6300A
- MOULD CASE CIRCUIT BREAKER (ABB SACE) UP TO 1600A
- ISOLATOR (ABB Control Oy)* UP TO 3150A
- FUES SWITCH (ABB Control Oy)* UP TO 800A

FUNCTIONAL DEVICES:

- AUTOMATIC TRANSFER SWITCH (ABB SACE) UP TO 6300A
- MOTOR STARTER (ABB Control Oy)* UP TO 1000KW

MEASURING AND INSTRUMENT METERS:

- AMMETER, VOLTMETER, KILOWATT HOUR METER, MAXIMUM DEMAND METER, SYNCHRONIZING METER, POWER FACTOR METER, I.D.M.T OVERCURRENT AND EARTH FAULT RELAYS...ETC.

Suitable partitions and segregations are provided for proper separation of switchgears and busbar to suit the Form 2, Form 3 and Form 4 requirement of the latest BS EN 60439 and IEC 60439, part 1 and other international standards. Standard production is Form 3b Type 2. It can be built according to the client's requirement.

For more information about L.V. switchgears, please contact the ABB local offices.

所有开关、计量和功能元件均安装在多个小室内。

开关：

- ABB - SACE 空气断路器：至 6300A
- ABB - S 系列塑壳断路器：至 1600A
- ABB - 隔离开关：至 3150A
- ABB - 熔断器开关：至 800A

功能件：

- ABB - 自动转换开关：至 6300A
- ABB - 马达开关：至 1000KW

仪表计量：

- 电压表、电流表、有功电表、最大需量表、同步表、功率因素表、I.D.M.T 过电流、接地故障继电器等。

按最新 IEC 60439 及 BS EN 60439 第一部份及其他国际标准，开关柜可分为 2 型、3 型和 4 型三种，标准生产为型式 3b，并按照合约的要求而提供适合的型式。

要详细了解 ABB 低压的产品系列，请与 ABB 地区办事处联系。

COMPARTMENT SIZE OF 'ABB SACE' AIR CIRCUIT BREAKER

ABB - 空气断路器模数要求

MODEL/TYPE 开关型号	NO. OF POLES 极	RATING 额定电流	MODULE SIZE (W x H) 模数(宽×高)	
			with indication lamps and meters 带指示灯及仪表	without indication lamps and meters 不带指示灯及仪表
F1S 800 / E1B08	3P	800A	4 x 4	3 x 3
	4P	800A	4 x 4	4 x 3
F1S 1250 / E2N12 / E1B12	3P	1250A	4 x 4	3 x 3
	4P	1250A	4 x 4	4 x 3
F1S 2000 / E2N20	3P	2000A	4 x 4	4 x 3
	4P	2000A	4 x 4	4 x 3
F2S 2500 / E3N25	3P	2500A	4 x 4	4 x 3
	4P	2500A	4 x 4	4 x 3
F4S 3200 / E3N32	3P	3200A	4 x 5 (For F4S 3200) / 4 x 4 (For E3N32)	4 x 4
	4P	3200A	4 x 5 (For F4S 3200) / 4 x 4 (For E3N32)	4 x 4

* Note: The Name 'ABB Control Oy' which is previously named as 'ABB Stromberg'

SWITCHGEAR COMPARTMENT

开关柜小室

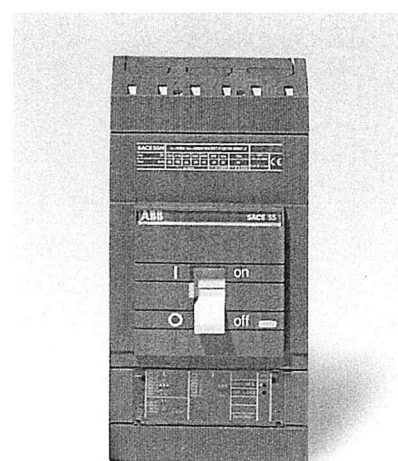
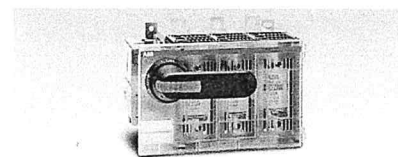
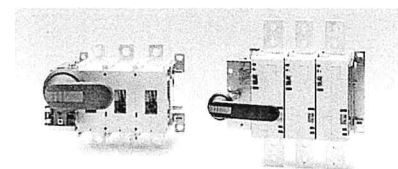


ABB 'SACE' MCCB Isomax
低压塑壳断路器 S 系列



'ABB Control Oy' rear connected fuse switch
ABB 后接式熔断器



'ABB Control Oy' front connected isolator
ABB 前接式隔离开关

ABB SACE 3P/4P REAR CONNECTED MOULDED CASE CIRCUIT BREAKER 三极/四极后接式塑壳断路器

MODEL/TYPE 开关型号	RATING 额定电流	3P/4P Module (W x H) 三极/四极模数(宽×高)	
		Vertical 直置式	Horizontal 横置式
S3	32A - 100A	2 x 2	2 x 2
S3	50A - 125A	2 x 2	2 x 2
S3 / S4	160A	2 x 2 / 2 x 3	2 x 2 / 3 x 2
S3 / S4	200 - 250A	2 x 2 / 2 x 3	2 x 2 / 3 x 2
S5	400A	2 x 3	3 x 2
S6	500 - 630A	3 x 3	3 x 3
S6	800A	3 x 3	3 x 3
S7	1250A	3 x 4	4 x 3
S7	1600A	3 x 4	4 x 3

The S3 model is thermomagnetic device protection.
S3 为热敏电磁保护。

'ABB Control Oy' 3P/4P REAR CONNECTED FUSE SWITCH 'ABB' 三极/四极后接式熔断器开关

MODEL/TYPE 开关型号	RATING 额定电流	3P/4P MODULE (WxH) 三极/四极模数(宽×高)
OESA 32G1	16A, 20A, 32A	2 x 2
OESA 63G1	40A, 63A	2 x 2
OESA 100G1	80A, 100A	2 x 2
OESA 160B3	125A, 160A	2 x 2
OESA 200B3	200A	2 x 2
OESA 250B3	250A	2 x 2
OESA 315B3	315A	3 x 2
OESA 400B3	250A, 315A, 400A	3 x 2
OESA 630B3	500A, 630A	3 x 3
OESA 800B3	800A	3 x 3

DIRECT ON LINE STARTER WITH REAR CONNECTED FUSES SWITCH OR MCCB (AC3 400V LL) 直接启动马达开关带后接式熔断器开关 或塑壳断路器 (AC3 400 伏)

KW rating 功率	I _r 电流	MCCB rating 塑壳断路器	F/S Model 熔断开关	Module (W x H) 模数(宽×高)	
				MCCB	F/S
below 3.7kw		16A	30A	2 x 2	2 x 2
3.7 - 10kw		30A	30A	2 x 2	2 x 2
11	22	30	30	2 x 2	3 x 2
15	30	30	30	2 x 2	3 x 2
18.5	37	40	60	2 x 2	3 x 2
22	44	63	80	3 x 2	3 x 2
30	60	80	80	3 x 2	3 x 3
37	72	100	100	3 x 2	3 x 3

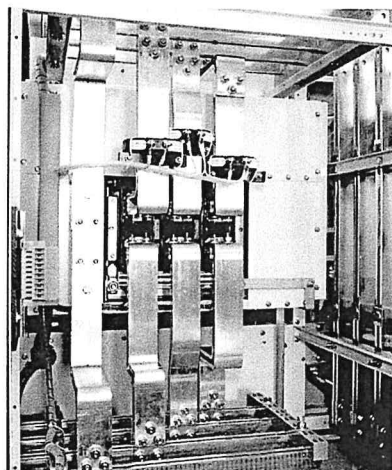
STAR DELTA STARTER WITH REAR CONNECTED FUSES SWITCH OR MCCB (AC3 400V LL) 星三角启动马达开关带后接式熔断器开关 或四极塑壳断路器 (AC3 400 伏)

KW rating 功率	I _r 电流	MCCB rating 塑壳断路器	F/S Model 熔断开关	Module (W x H) 模数(宽×高)	
				MCCB	F/S
11	22	30	30	3 x 2	3 x 2
15	30	30	30	3 x 2	3 x 2
18.5	37	40	60	3 x 2	3 x 3
22	44	60	80	3 x 2	3 x 3
30	60	80	80	3 x 2	4 x 3
37	72	100	100	4 x 3	4 x 3
45	85	100	100	4 x 3	4 x 3
55	105	125	125	4 x 3	4 x 3

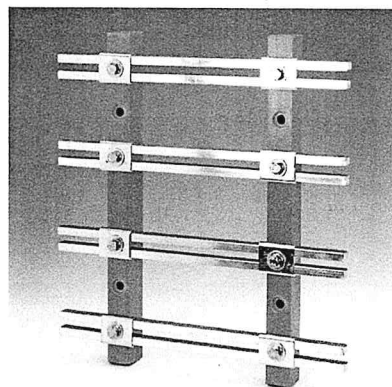
Alternative starting methods such as Auto-TX starting and closed type transition starting are also available on request.
其它型式的马达启动方法，如自耦式变压器及闭合式星三角启动器，也可以提供。

BUSBAR COMPARTMENT

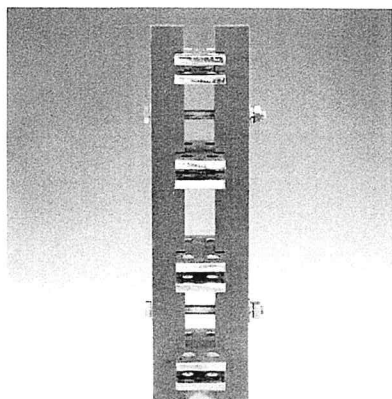
母线小室



Sectional ACB with upper and lower Busbar connected.
母联 ACB 与上下母线的连接。



800A ~ 1250A Busbar Support.
800A ~ 1250A 母线。



1600A and above Busbar support.
1600A 以上的母线。

The busbar system of the **ABB MD 190 (HONOR)** Switchboard is located behind switchgear compartment to facilitate the connection of busbar to the switchgears. All busbars provided are hard drawn, high conductivity, electro tin-plated copper to BS 1432 or BS 1433.

The busbar may be fixed at the top, bottom or middle part of the switchboard. The horizontal main busbars and vertical busbars are supported by the moulded type glass fibre reinforced polyester which have high mechanical strength, good heat resistance and water resistance.

The standard thickness of main busbar is 10mm with two conductors running in parallel per phase. The system can be suitable for extension at both end of the switchboard to cater for the future extension of the switchboard on site.

ABB MD 190 (HONOR) ‘安亚’ 系列开关柜母线系统布置在柜内的开关小室后面方便连接。所有母线均按 BS 标准，并采用冷拉和镀锡加工，提高防霉性能。

母线可安装在柜顶部，底部或中部。水平主母线和垂直分支母线用玻璃纤维加强型聚脂排夹件固定。具有高机械强度，抗温升和防水功能。

主母线的标准厚度为 10mm，每相双拼。该母线系统适合将来在现场扩展连接时采用。

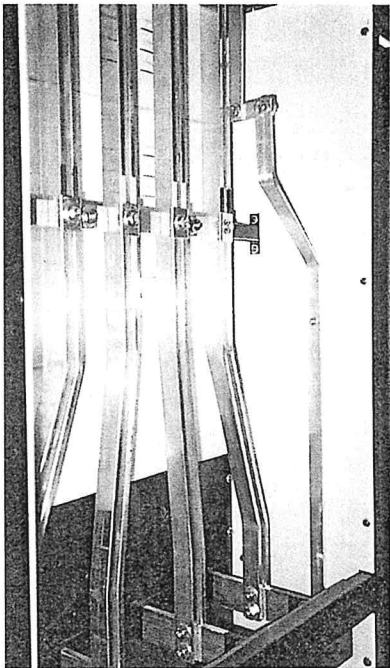
HORIZONTAL / VERTICAL MAIN BUSBAR TABLE

水平及垂直分支母线表

RATING 电流	COPPER BUSBAR SIZE 铜材母线规格
3200A	2 x 100 x 10 mm
2500A	2 x 75 x 10 mm
2000A	2 x 63 x 10 mm
1600A	2 x 50 x 10 mm
1250A	2 x 40 x 10 mm
1000A	2 x 30 x 10 mm
800A	2 x 20 x 10 mm / 1 x 40 x 10 mm
630A	1 x 30 x 10 mm
400A	1 x 30 x 6 mm

CABLE COMPARTMENT

电缆小室



Busbar connect to Transformer.
由开关柜连接至变压器的母线。

The cable compartment or rear access type switchboard is located at the back of the switchboard where all the switchgear's outgoing terminals and incoming cables/busbars are terminated for connections. The safety partitions between the busbar compartment and cable compartment are segregated by louvered plates to facilitate ventilation of heat. At the lower part of the switch-board, there is an earth busbar of standard size 1x25x6mm which is running on the whole length of the switchboard. Alternative earthing bar size may be available to suit the site requirement.

The Cable compartment of front access type switchboard is located adjacent to the switchgear compartment. The front access type switchboard is suitable to BS EN60439 pt1 Form 4 Type 2 requirement. The back to back arrangement of front access switchboard can optimize the switch-room space. While for the rear access design, Form 4 Type 5 and 6 is our standard design.

Suitable cable trays is provided as option for securing the incoming and outgoing cables. The switchboard is capable for both top and bottom incoming and outgoing of the busduct (top only) and cables.

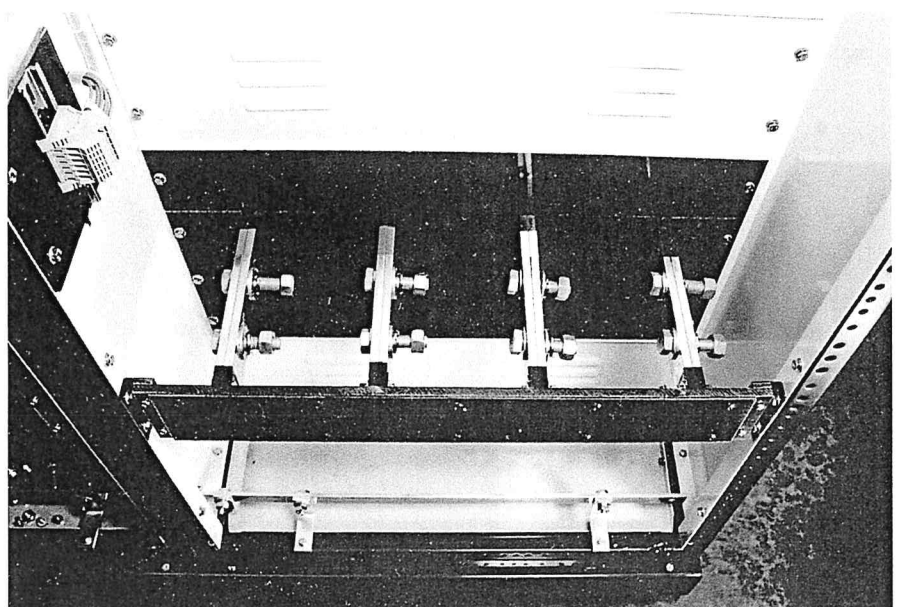
Additional ventilation fans can be fixed at the back or both sides of the switchboard as options for IP31 design.

后出线式的电缆小室布置在开关柜的背部。在柜的背部排列着所有的开关小室的进出线或母线的接线端子。母线小室和电缆小室设有安全隔板。隔板带网孔便于散热。在柜的底部有标准的接地排 1x25x6mm，水平式贯穿开关柜内。其它截面的接地排，也可按要求供货。

前出线式的电缆小室布置在开关的左右两旁。前出线式适合 IEC 60439, BS EN60439 第一部份 4 型的设计。两个背靠背前操式开关柜可放置在配电房的中间，故能有效地使用配电房的空间。

ABB 公司还可提供进出线电缆接线架。开关柜母线和电缆进出线可任选上或下进出（但出线钢排只能向上出）。

IP31 防护等级开关柜可在柜背部或两侧装设冷却风机。



Cabling compartment suitable for cable and busbar coming from top or bottom.
方便底顶出线的母线小室。

TYPICAL ARRANGEMENT

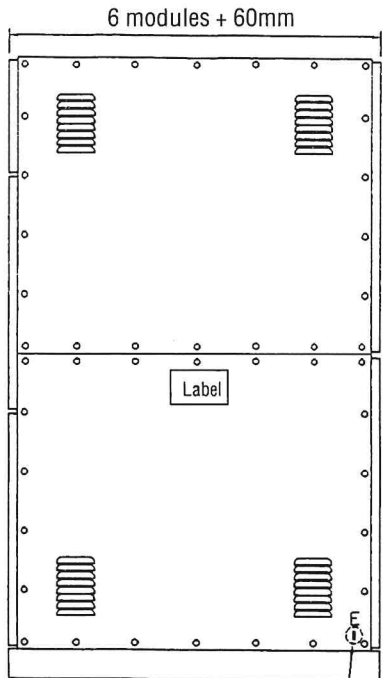
单元组合型式示例

TYPICAL ARRANGEMENT 1

单元组合示例 1

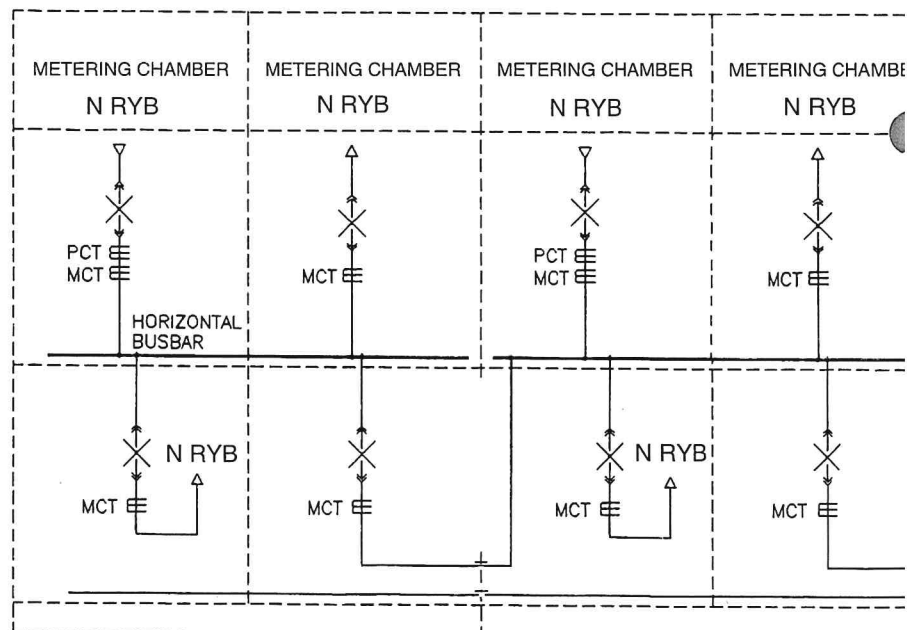
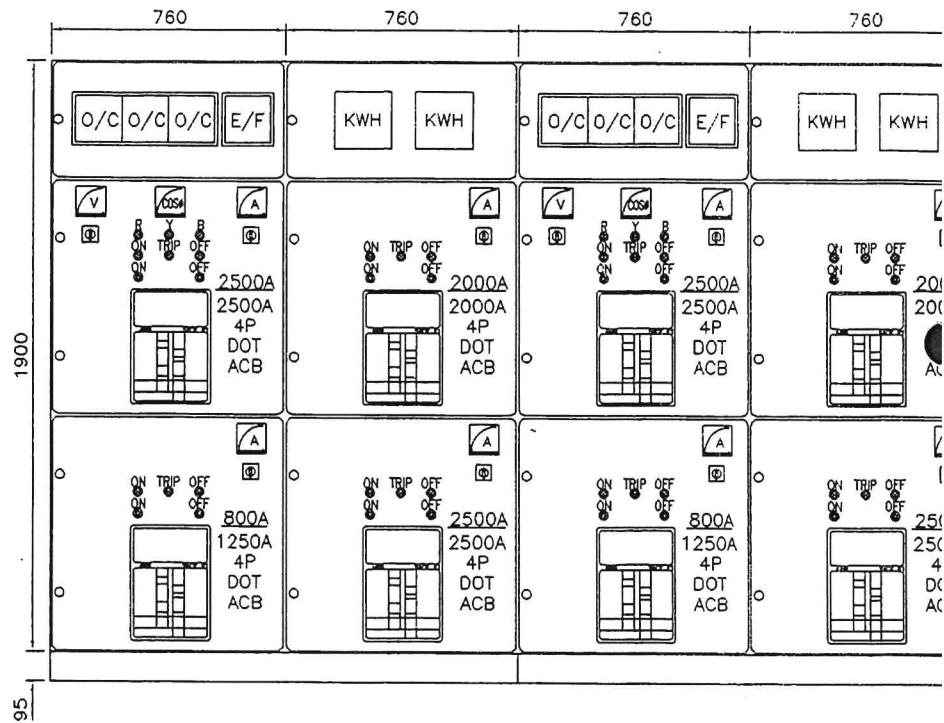
TYPICAL FORM 3b TYPE 2 SWITCHBOARD WITH

典型型式 3b 是后接式开关柜

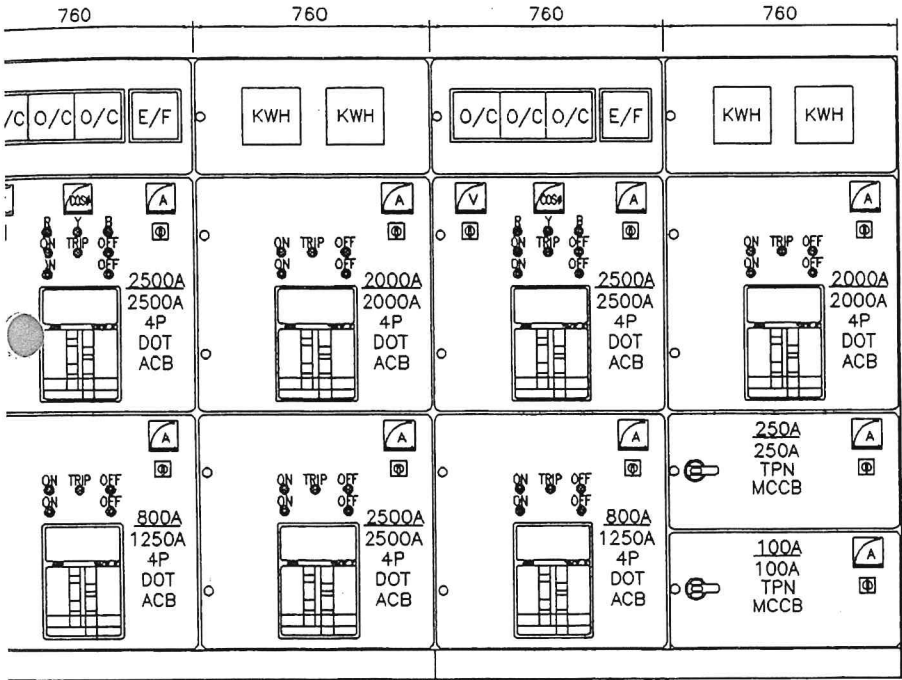


SIDE VIEW
侧面图

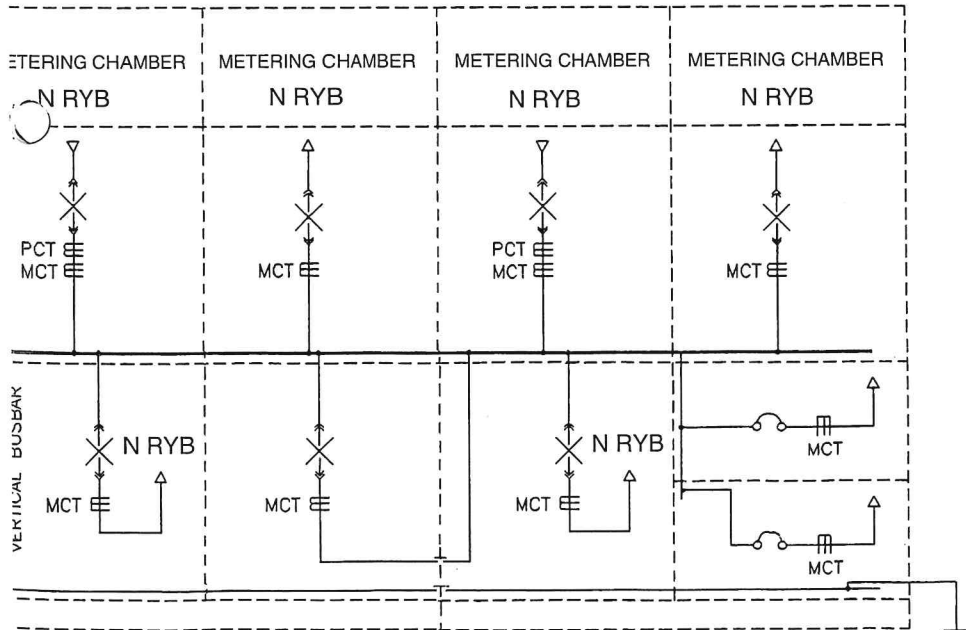
EARTHING
TERMINAL
接地排



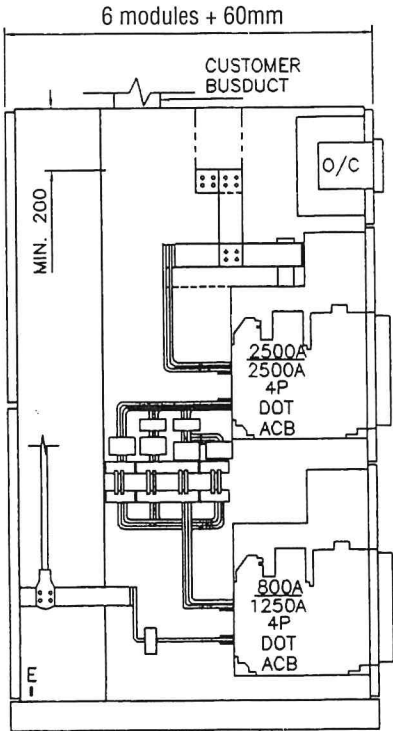
AND BOTTOM REAR ACCESS CABLE AND BUSBAR 上位及下位进出电缆与密封母排



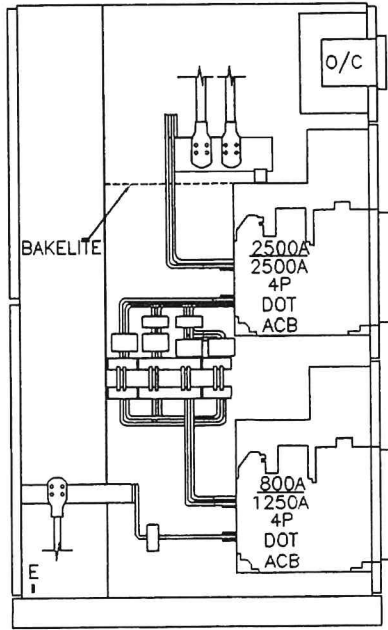
FRONT VIEW
正视图



BUSBAR ARRANGEMENT
母线排位



SECTIONAL VIEW FOR CABLE AND
BUSBAR ENTRY FROM TOP
剖视图见密封母排及电缆接法



SECTIONAL VIEW FOR CABLE ENTRY FROM
TOP AND BOTTOM
剖视图见上位及下位进出电缆接法